

Biochemistry and Physiology of Marine Animals

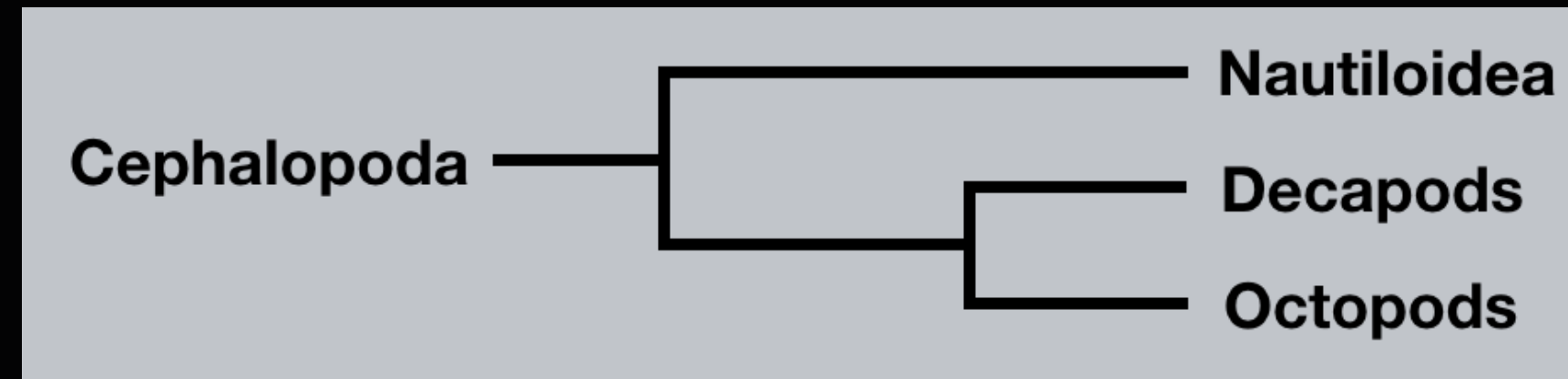
Cephalopods' adaptations to their environment

Focus on the *Hapalochlaena* genus

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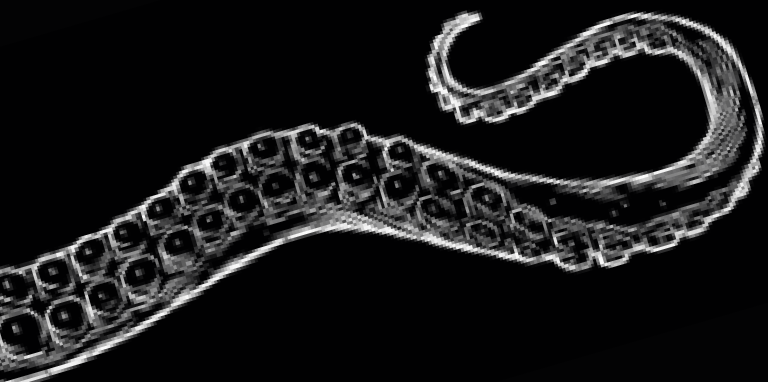


I. Origin of the genus

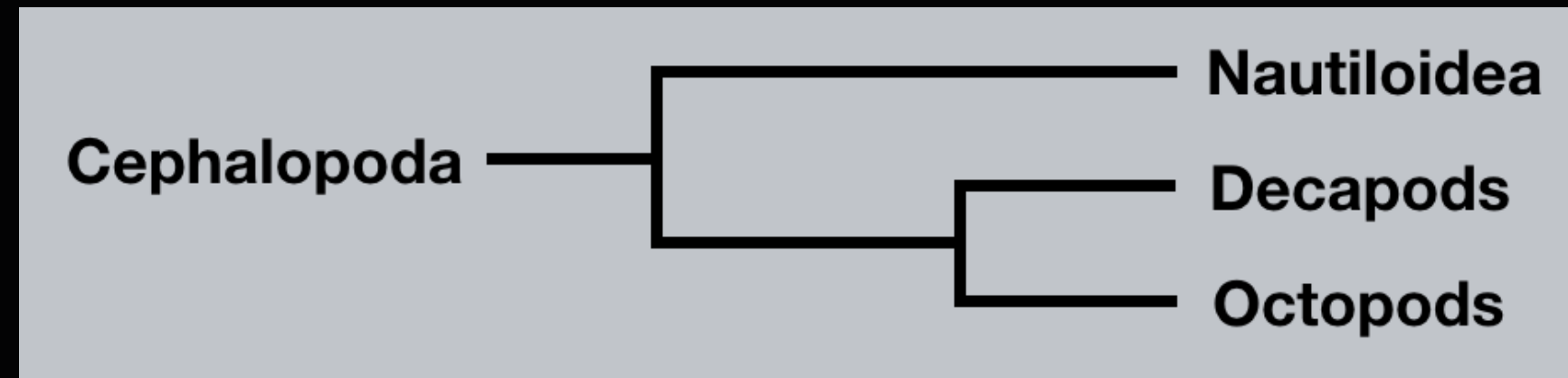


Cephalopods :

- Bilateral body symmetry
- Set of 'arms'
- Well-developed nervous system



I. Origin of the genus



Nautiloidea :

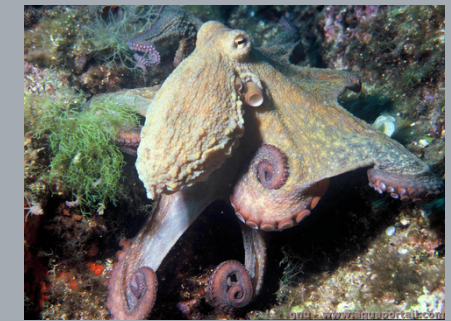
→ External shell



Coleoidea :

→ Decapods : 10 appendages (8 arms & 2 elongated tentacles)

→ Octopods : 8 appendages



I. Origin of the genus

The genus *Hapalochlaena*

- Divergence around 59 million years ago (Eocene-Oligocene extinction)
- Molecular studies support the monophyletic status
- 4 described species

H. maculosa



H. lunulata



H. fasciata



H. nierstraszi

One single
specimen
observed in 1938

BUT Molecular data suggests more clades

II. Distribution area

- **Pacific and Indian**
- Highest diversity at the equator
- Temperature range : 20-28°C
- Depth : up to 50m
- Rocky reefs



Figure 1: Habitat of the blue ringed octopus

H. lununata : **merobenthic**

H. maculosa & *H. fasciata* : **holobenthic**



III. Reproductive Behaviour

4.1 Mating Behavior



Complex behaviors: color changes, physical displays, and mounting techniques

- **Polygamous**
- Mating occurs in **spring/autumn**

Mating process : Male mounts female, transfers spermatophores via **hectocotylus** (specialized reproductive arm)

Duration : Several **minutes** to **hours** in protected areas to avoid predation



Figure 2: Copulation of *H. maculosa* (Tranter & Augustine, 1973)

III. Reproductive Behaviour

4.2 Social Recognition and Partner Choice

Octopuses = **asocial**, interactions mostly for reproduction
→ **Social recognition (mate choice & sperm competition)**



Sex Recognition

- Females = **distance chemoreception**
- Males = **physical contact**

Mating History.

- Males adjust **copulation duration** based on **sperm competition**
- Females can store sperm from **multiple males** and produce a **single clutch** with **mixed paternity**

Aggressive post-mating behavior: avoidance, resumption, or aggression (cannibalism)

III. Reproductive Behaviour

4.3 Egg Laying



- **50-100** eggs in **secure areas** (reefs, burrows)
- Eggs wrapped in a **brood pouch** formed by female arms
- Brooding last **~2 month**

- **Egg development** varies by species and temperature (*H. lunulata* : **~4-5 weeks**)
- **Blue rings** appear **~ 6 weeks** post-hatching



Figure 3 : Development of *H. lunulata* eggs
(Williams & Caldwell, 2024)

- Female **Stop feeding**, focusing on egg care
- Lifespan : **~ 1 year** reproductive phase mark the end → **Senescence**

IV. Camouflage

Specialized cells

Chromatophores :

- Sac filled with pigments
 - Brown, black, red, yellow
 - Neuro-muscular control
- Leucophores and iridophores
- Large range of body patterns
 - Camouflage
 - Communication

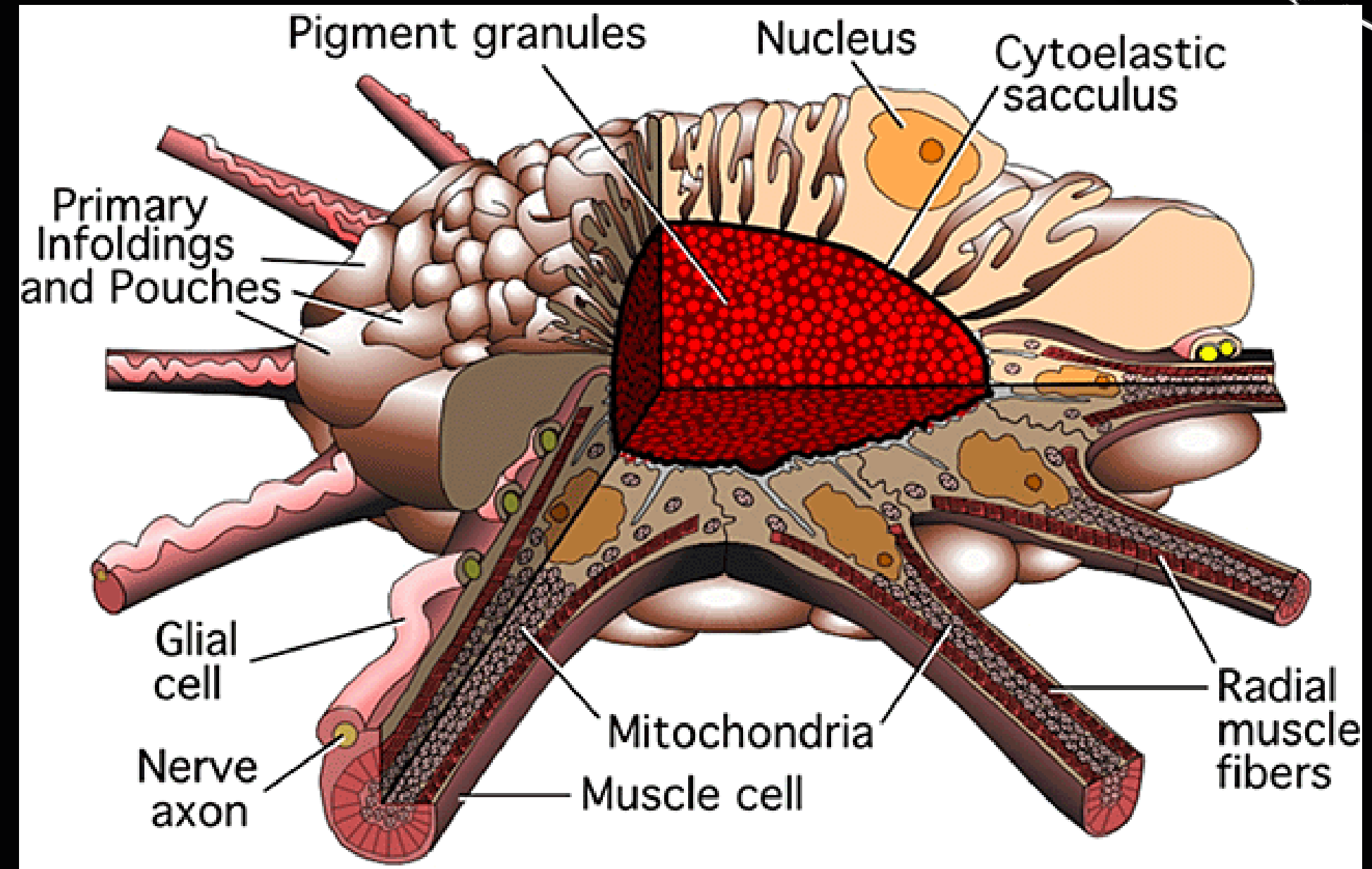
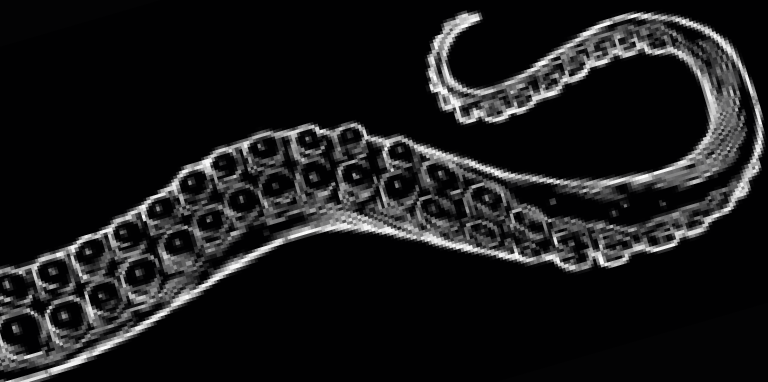


Figure 4: Diagram of a retracted cephalopod chromatophore (Messenger, 2001).

IV. Camouflage

Camouflage control

- Intensity of the iridescence
 - Muscarinic cholinergic receptors
 - Take few minutes
- Block and filter the iridescence
 - Via muscle, **neuronal control**
 - Chromatophores
 - Almost **instant**



Blue rings

- Iridophores
 - Controlled by **nervous system**
- Hide the reflectors
- < 0.3 seconds
- Chromatophores **accentuate** the **contrast**

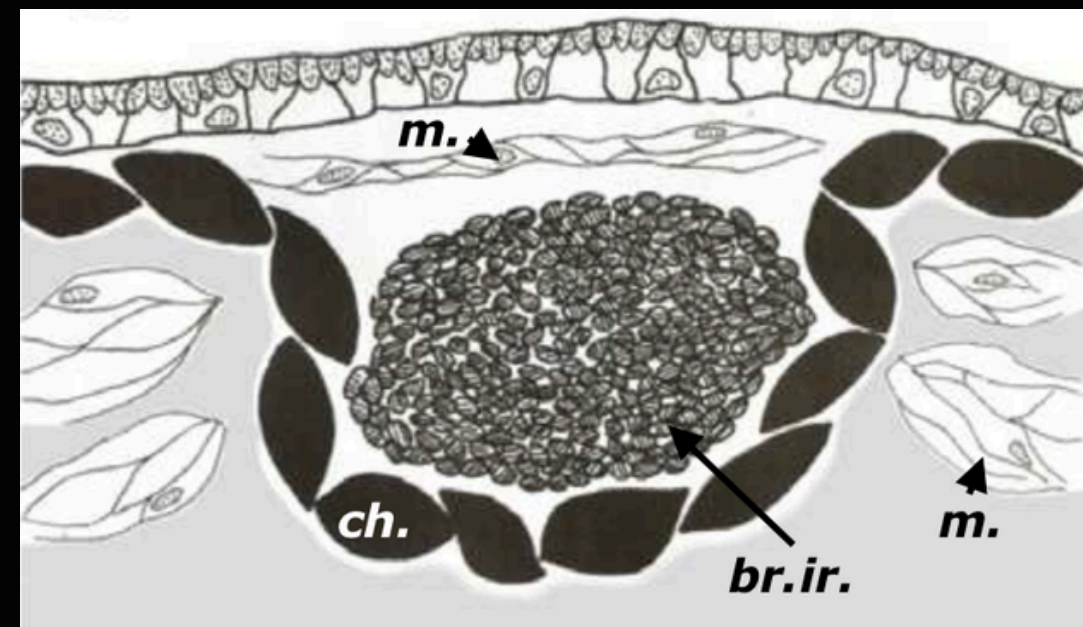


Figure 5 : Diagram of the opening (left) of the blue rings (Mäthger et al., 2012) and picture of a the blue rings

V. Toxine

- **Tetrodotoxin**
- Not a protein venom $\gg$ *other octopuses*
- Stored in the **posterior salivary gland**

- Voltage-dependent **Na channels**
 - Lead to muscle **paralysis**
- **Mutation** in Na channels loops
 - **Resistance**
- Binding proteins to transport TTX

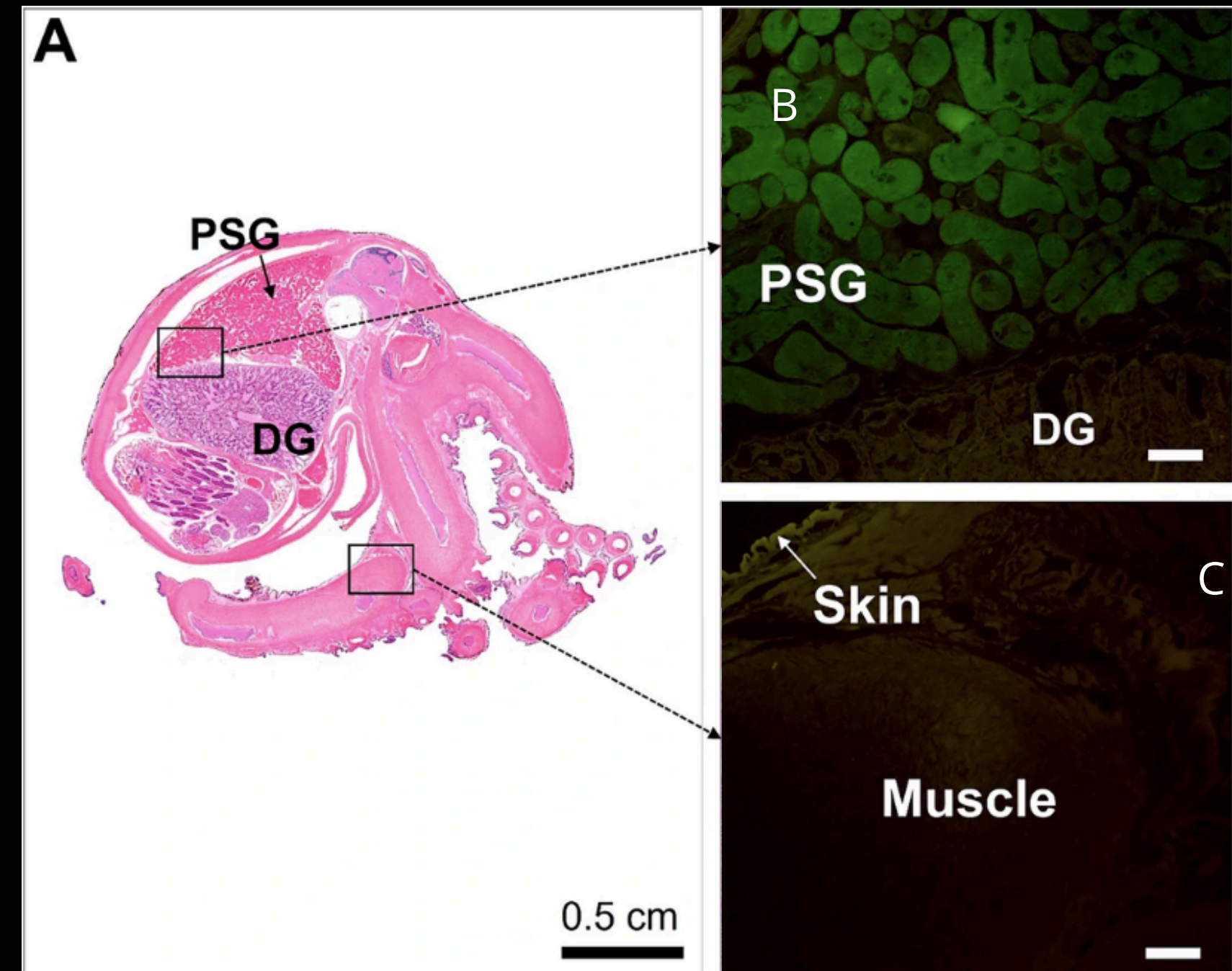
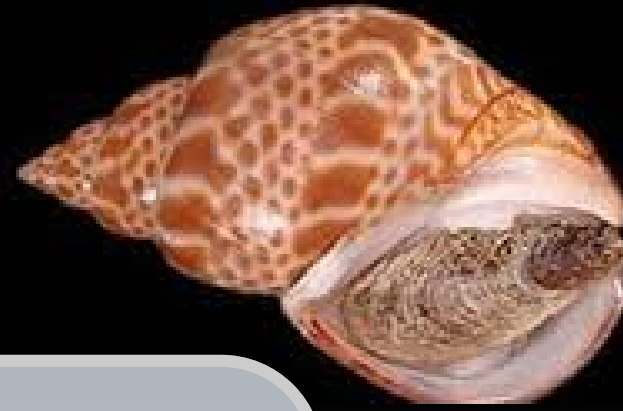


Figure 6 : Longitudinal section of *H. fasciata* stained with H&E, (A) TTX-specific fluorescence in the PSG ; (C) lack of TTX-specific fluorescent in the arm skin and muscle (Kajino et al., 2024)

V. Toxine



Origin of TTX

- Bacterial via **ingestion** or **endosymbiosis**
 - Protobacteria *Pseudomonas* (41%)
 - *Bacillus* (21%)



Role of TTX

- **Hunting** as injected venom
 - PSG
- **Defense** as ingested poison
 - Skin, ink, muscle

In other organisms

- **Marine** and **terrestrial** species
- 60 species in 8 phyla
- **Bacterial** origin
- In all **TTX-resistant** species
 - mutation in TTX-binding site in Na channel

Conclusion

- **Soft bodies + camouflage** = perfectly blending in their environment
- **Blue rings** = visual signals
 - To scare predators
 - Increase their survival rate
- **Tetrodotoxin**
 - Hunt and defense
- **Reproductive strategies** ensure the **sustainability** of the species

Each morphological, physiological and behavioral trait responds precisely to the requirements of their environment.

Perspectives

→ Lack of knowledge

- Need of **molecular analysis** to determine the species (expansive)
 - Really **difficult** to **cultivate** the TTX-producing bacteria
 - TTX **biosynthesis pathway** in *Hapalochlaena* ?
 - **High health risk** : more and more poisoned people
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- Very **hard to observe**
 - Small size, lives in crevices and camouflages itself very well
 - Strong morphological resemblance between the different species
 - **Identification difficult**
 - **Deadly** : dangerous handling
 - **Short life** cycle and difficulty in keeping them in captivity



Thank you for your
attention !

Some questions ?